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Unit – 3 : Session – 5 : SLO – 1 | Infix to Postfix Conversion — Worked Example

■ I. Step-by-Step Trace: $A - (B / C + (D \% E * F) / G) * H$

- A -> operand | Output: A | Stack: -
- - -> push | Output: A | Stack: -
- (-> push | Output: A | Stack: - (
- B -> operand | Output: AB | Stack: - (
- / -> push | Output: AB | Stack: - (/
- C -> operand | Output: ABC | Stack: - (/
- + -> pop /, push + | Output: ABC/ | Stack: - (+
- (-> push | Output: ABC/ | Stack: - (+ (
- D -> operand | Output: ABC/D | Stack: - (+ (
- % -> push | Output: ABC/D | Stack: - (+ (%
- E -> operand | Output: ABC/DE | Stack: - (+ (%
- * -> pop %, push * | Output: ABC/DE% | Stack: - (+ (*
- F -> operand | Output: ABC/DE%F | Stack: - (+ (*
-) -> pop till (| Output: ABC/DE%F* | Stack: - (+
- / -> pop +, push / | Output: ABC/DE%F*+ | Stack: - (/
- G -> operand | Output: ABC/DE%F*+G | Stack: - (/
-) -> pop till (| Output: ABC/DE%F*+G/ | Stack: -
- * -> push | Output: ABC/DE%F*+G/ | Stack: - *
- H -> operand | Output: ABC/DE%F*+G/H | Stack: - *
- END -> pop *, pop - | Output: ABC/DE%F*+G/H*- | Stack: empty

■ II. Final Answer

- Postfix expression: $A B C / D E \% F * G / + H * -$
- Compact form: $ABC/DE\%F*+G/H*-$